Lecture Five

1 Linear Independence

1.1 Introduction

The homogeneous equations can be studied from a different perspective by writing them as a vector equation. In this way the focus shifts from the unknown solutions of $Ax = 0$ to the vectors that appear in the vector equations. For instance, consider the equation

$$x_1 \begin{bmatrix}1\\2\\3\end{bmatrix} + x_2 \begin{bmatrix}4\\5\\6\end{bmatrix} + x_3 \begin{bmatrix}2\\1\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} \tag{1.1.1}$$

This equation has a trivial solution of course, where $x_1 = x_2 = x_3 = 0$. The main issue is whether the trivial solution is the only one.

Definition 1.1.1 *An indexed set of vectors $\{v_1, v_2, \dots, v_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation*

$$x_1v_1 + x_2v_2 + \cdots + x_pv_p = 0$$

*has only the trivial solution. The set $\{v_1, v_2, \dots, v_p\}$ is said to be **linearly dependent** if there exist weights $c_1, c_2, \dots, c_p$ not all zero, such that*

$$c_1v_1 + c_2v_2 + \cdots + c_pv_p = 0 \tag{1.1.2}$$

Equation 1.1.2 is called a **linear dependence relation** among $v_1, v_2, \dots, v_p$ when the weights are not all zero.

Example 1.1.1 *Let* $v_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}$, $v_2 = \begin{bmatrix}4\\5\\6\end{bmatrix}$ *and* $v_3 = \begin{bmatrix}2\\1\\0\end{bmatrix}$

(a) Determine if the set $\{v_1, v_2, v_3\}$ is linearly independent.

(b) If possible, find a linear dependence relation among $\{v_1, v_2, v_3\}$.

Sol

(a) We must determine if there is a non trivial solution or not. Row operations on the associated augmented matrix show that:

$$\begin{bmatrix}1&4&2&0\\2&5&1&0\\3&6&0&0\end{bmatrix} \to \begin{bmatrix}1&4&3&0\\0&-3&-3&0\\0&0&0&0\end{bmatrix}$$

Clearly x_1 and x_2 are basic variables and x_3 is free. Thus each non zero value of x_3 determines a non trivial solution of equation 1.1.2. Hence v_1, v_2, v_3 are linearly dependent (and not linearly independent).

(b) To find a linear dependence relation among v_1, v_2 and v_3 completely row reduce the augmented matrix and write the new system.

$$\begin{bmatrix}1&0&-2&0\\0&1&1&0\\0&0&0&0\end{bmatrix} \quad \begin{aligned} x_1 - 2x_3 &= 0\\ x_2 + x_3 &= 0\\ 0 &= 0\end{aligned}$$

Thus $x_1 = 2x_3$, $x_2 = -x_3$ and x_3 is free. Choose any non zero value for x_3 say $x_3 = 5$, then $x_1 = 10$ and $x_2 = -5$. Substitute these values into equation 1.1.1 obtain

$$10v_1 - 5v_2 + 5v_3 = 0$$

This one (out of infinitely many) possible linear dependence relations among v_1, v_2 and v_3.

1.2 Linear Independence of Matrix Columns

Suppose that we begin with a matrix $A = [a_1 \ \dots \ a_n]$ instead of a set of vectors. The matrix equation $Ax = 0$ can be written as:

$$x_1a_1 + x_2a_2 + \cdots + x_na_n = 0$$

Each linear dependence relation among the columns of A corresponds to a non trivial solution of $Ax = 0$. And here we have the following fact.

The columns of a matrix A are linearly independent iff the equation Ax=0 has only the trivial solution

Example 1.2.2 *Determine if the columns of the matrix* $A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & -1 \\ 5 & 8 & 0 \end{bmatrix}$ *are linearly independent*

Sol

To study $Ax = 0$ *row reduce the augmented matrix*

$$\begin{bmatrix} 0 & 1 & 4 & 0 \\ 1 & 2 & -1 & 0 \\ 5 & 8 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & -2 & 5 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 13 & 0 \end{bmatrix}$$

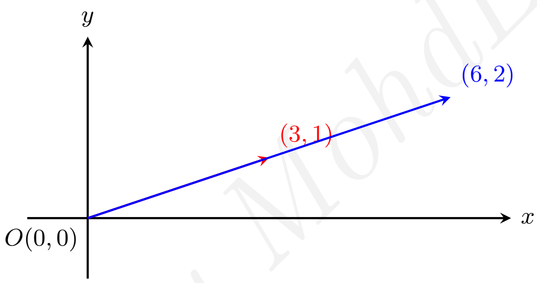
At this point, it is clear that there are three basic variables and no free variables. So the equation $Ax = 0$ *has only the trivial solution, and the columns of* A *are linearly independent.*

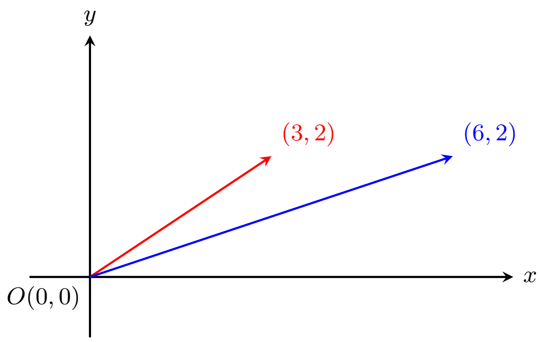
Example 1.2.3 *Determine if the following sets of vectors are linearly independent.*

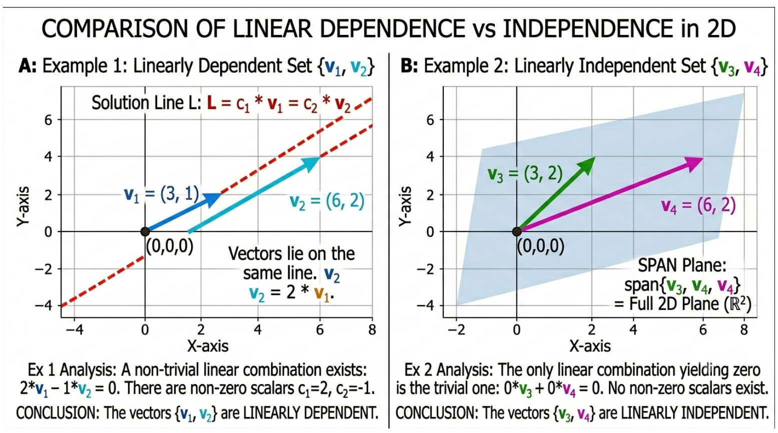
1. $v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$
2. $v_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$

Sol

1. *Notice that* v_2 *is a multiple of* v_1*, namely* $v_2 = 2v_1$*. Hence* $-2v_1 + v_2 = 0$ *which shows that* $\{v_1, v_2\}$ *is linearly dependent.*
2. *The vectors* v_1 *and* v_2 *are certainly not multiples of one another. Could they be linearly dependent? Suppose* c *and* d *satisfy*
$$cv_1 + dv_2 = 0$$
If $c \neq 0$*, then we can solve for* v_1 *in terms of* v_2*, namely* $v_1 = \left(\frac{d}{c}\right) v_2$*. This result is impossible because* v_1 *is not a multiple of* v_2*, So* c *must be zero. Similarly* d *also must be zero. Thus* $\{v_1, v_2\}$ *is a linearly independent set.*







Theorem 1.2.1 ***Characterization of Linearly Dependent Sets:*** *An indexed set $S = \{v_1, \dots, v_p\}$ of two or more vectors is linearly dependent if and only if at least on e of the vectors in S is a linear combination of the others. In fact if S is linearly dependent and $v_1 \neq 0$, then some v_j (with $j > 1$) is a linear combination of the preceding vectors $v_1, \dots, v_{j-1}$.*

The next two theorems describes special cases in which the linear dependence of a set is automatic.

Theorem 1.2.2 *If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.*

Example 1.2.4 *The vectors $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 4 \\ -1 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$ are linearly dependent by theorem 1.2.2 because there are three vectors in the set and there are only two entries in each vector.*

Theorem 1.2.3 *If a set $S = \{v_1, \dots, v_p\}$ in $\mathbb{R}^n$ contains the zero vector, then the set is linearly dependent.*

Example 1.2.5 *Determine by inspection if the given set is linearly dependent.*

(a) $\begin{bmatrix} 1 \\ 7 \\ 6 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 0 \\ 9 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}$, $\begin{bmatrix} 4 \\ 1 \\ 8 \end{bmatrix}$

(b) $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \\ 8 \end{bmatrix}$

(c) $\begin{bmatrix} -2 \\ 4 \\ 6 \\ 10 \end{bmatrix}$, $\begin{bmatrix} 3 \\ -6 \\ -9 \\ -15 \end{bmatrix}$

Sol

(a) The set contains four vectors, each of which has only three entries, so the set is linearly dependent by theorem 1.2.2

(b) Theorem 1.2.2 does not apply here because the number of the vectors does not exceed the number of the entries in each vector. Since the zero vector is in the set, the set is linearly dependent by theorem 1.2.3

(c) Compare the corresponding entries of the two vectors. The second vector seems to be $-\frac{3}{2}$ times the first vector. This relation holds for the first three pairs of entries, but fails for the fourth pair. Thus neither of the vectors is a multiple of the other, and hence they are linearly independent.

Exercise 1.2.1 *Let $u = \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}$, $v = \begin{bmatrix} -6 \\ 1 \\ 7 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}$ and $z = \begin{bmatrix} 3 \\ 7 \\ -5 \end{bmatrix}$*

Are the the sets $\{u, v\}, \{u, w\}, \{u, z\}, \{v, w\}, \{v, z\}$ and $\{w, z\}$ are linearly independent? Why or why not?

2 Linear Transformation

2.1 Introduction

In this section, for the equation $Ax = b$ we will think of the matrix A as an object that 'acts' on a vector x by multiplication to produce a new vector called Ax. For instance

$$\underbrace{\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 5 \\ 8 \end{bmatrix}}_{b}$$

and

$$\underbrace{\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 1 \\ 4 \\ -1 \\ 3 \end{bmatrix}}_{u} = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{0}$$

Say that multiplication by A **transform** x into b and **transform** u into the zero vector.

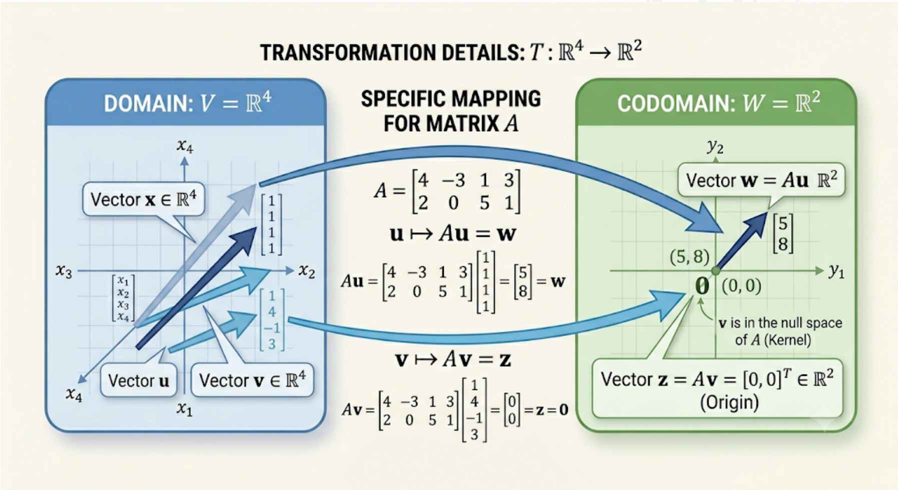


Figure 1: Linear transformation mapping vectors **u** and **v** using matrix A.

From this new point of view, solving the equation $Ax = b$ amounts to finding all vectors x in $\mathbb{R}^4$ that are transformed into the vector b in $\mathbb{R}^2$ under the 'action' of multiplication of A.

2.2 What is transformation?

Definition 2.2.2 *A **transformation** (or **function** or **mapping**) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector x in $\mathbb{R}^n$ a vector $T(x)$ in $\mathbb{R}^m$. The set $\mathbb{R}^n$ is called the **domain** of T, and $\mathbb{R}^m$ is called the **codomain** of T.*

Note:

1. The notation $T : \mathbb{R}^n \to \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$.
2. For $x \in \mathbb{R}^n$, the vector $T(x) \in \mathbb{R}^m$ is called the image of x under the action of T.
3. The set of all images $T(x)$ is called the range of T.
4. The range of T is the set of all linear combinations of columns of A because each image $T(x)$ is of the form Ax.

Example 2.2.6 *Let* $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$ *and* $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $c = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$. *and define a transformation* $T(x) = Ax$, *so that* $T(x) = Ax = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$

(a) *Find* $T(u)$ *the image of* u *under the transformation* T.

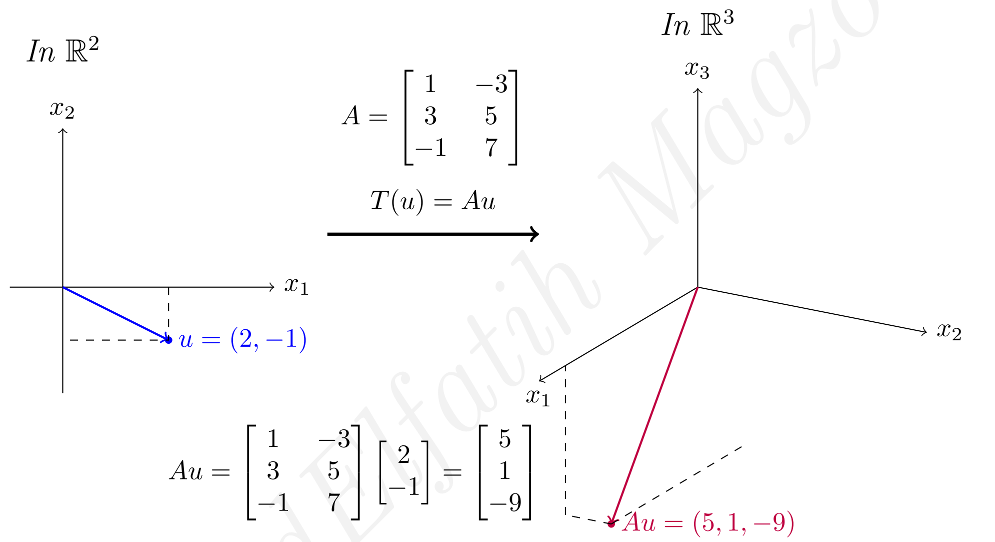
(b) *Find an* x *in* $\mathbb{R}^2$ *whose image under* T *is* b.

(c) *Is there more than one* x *whose image under* T *is* b?

(d) *Determine if* c *is in the range of the transformation* T.

Sol

(a) *Compute:* $T(u) = Au = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$.



(b) *Solve* $T(x) = b$ *for* x. *That is solve* $Ax = b$.

$$\begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix} \tag{2.2.1}$$

Row reduce the augmented matrix

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & -5 \end{bmatrix} \to \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & \frac{3}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \tag{2.2.2}$$

Here $x_1 = \dfrac{3}{2}, x_2 = -\dfrac{1}{2}$ *and* $x = \begin{bmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{bmatrix}$. *The image of this* x *under* T *is the given vector* b.

(c) *Any* x *whose image under* T *is* b *must satisfy equation 2.2.1. From equation 2.2.2, it is clear that equation 2.2.1 has a unique solution, so there is exactly one* x *whose image is* b.

(d) *The vector* c *is in the range of* T *if* c *is the image of some* x *in* $\mathbb{R}^2$, *that is if* $c = T(x)$ *for some* x. *This is just another way of asking if the system* $Ax = c$ *is consistent. To find the answer, row reduce the augmented matrix.*

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & 5 & 2 \\ -1 & 7 & 5 \end{bmatrix} \to \begin{bmatrix} 1 & -3 & 3 \\ 0 & 14 & -7 \\ 0 & 4 & 8 \end{bmatrix} \to \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 14 & -7 \end{bmatrix} \to \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -35 \end{bmatrix}$$

The third equation, $0 = -35$ *shows that the system is inconsistent, so* c *is not in the range of* T.

2.3 Concept of Linear Transformation

Definition 2.3.3 *A transformation (or mapping) T is said to be linear if*

(i) $T(u+v) = T(u) + T(v) \quad \forall u, v \in$ *domain of* T

(ii) $T(cu) = cT(u)$ *for all* $c \in \mathbb{R}$ *and* $u \in D(T)$

Property (i) says that the result $T(u+v)$ of first adding u and v in $\mathbb{R}^n$ and then applying T is the as first applying T to u and v and then adding $T(u)$ and $T(v)$ in $\mathbb{R}^m$. These two properties lead easily to the following useful facts:

1. If T is a linear transformation, then $T(0) = 0$
2. $T(cu + dv) = c\,T(u) + d\,T(v)$ for all vectors u, v in domain of T.

And we can generalize it as

$$T(c_1v_1 + c_2v_2 + \cdots + c_pv_p) = c_1T(v_1 + c_2T(v_2)) + \cdots + c_pT(v_p)$$

Example 2.3.7 *Define a linear transformation* $T : \mathbb{R}^2 \to \mathbb{R}^2$ *by*

$$T(x) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

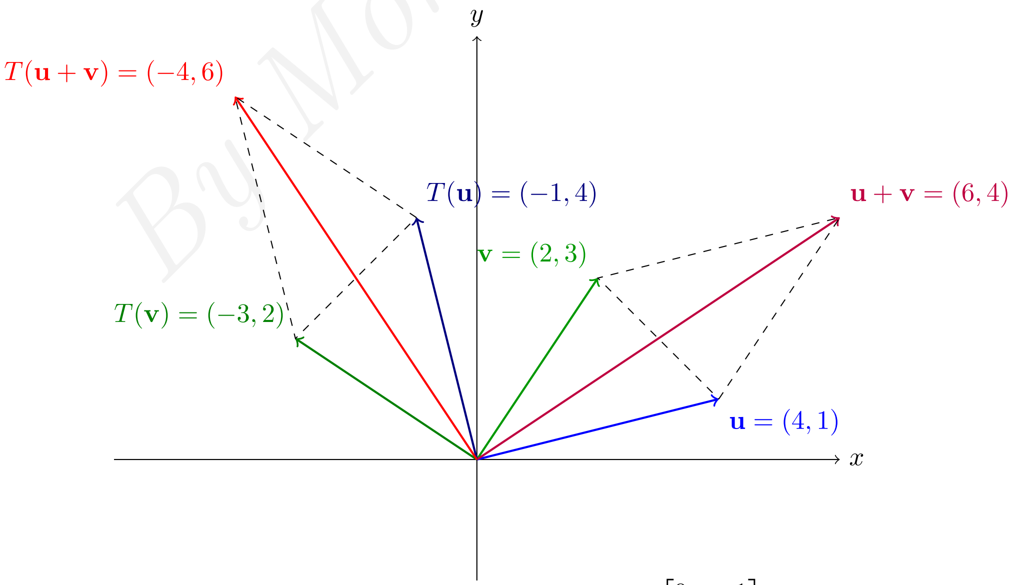
Find the images under T of $u = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ *and* $u + v = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$

Sol

$$T(u) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

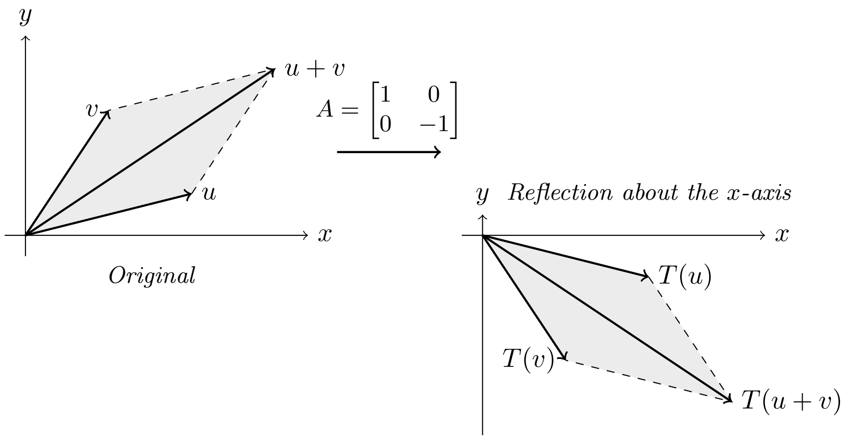
$$T(v) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

$$T(u) + T(v) = \begin{bmatrix} -4 \\ 6 \end{bmatrix} \quad \textit{and} \quad T(u+v) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

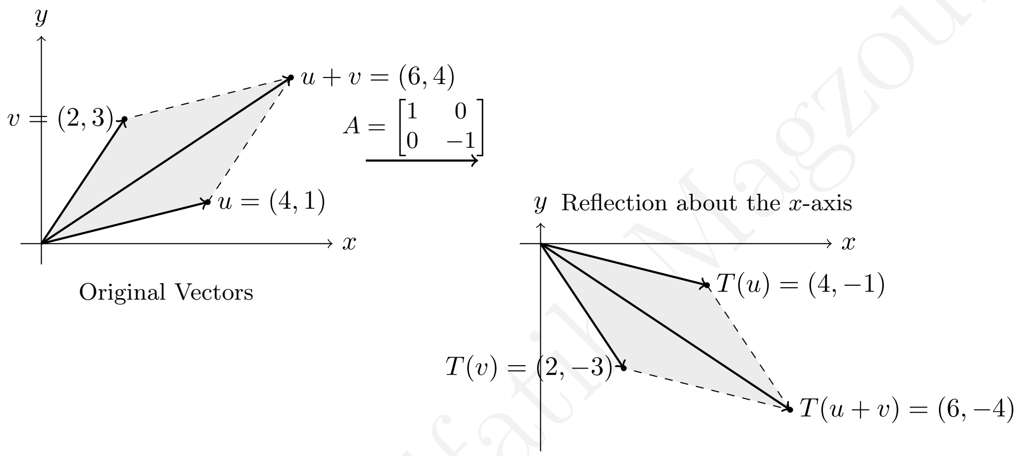


Linear Transformation $T(x) = Ax$, $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Example 2.3.8 *Let* $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. *Give a geometric description of the transformation* $x \to Ax$



By choosing the vectors u, v from the previous example we can see the description of the transformation as follows.



Exercise 2.3.2 *Suppose that $T : \mathbb{R}^5 \to \mathbb{R}^2$ and $T(x) = Ax$ for some matrix A and for each x in $\mathbb{R}^5$. How many rows and columns does A have?*